# SIMATS ENGINEERING
## ASSESSMENT TOOL 1: Concept Mapping

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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| Name | **P.Guru** |
| Register Number | **192521010** |

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**COURSE OUTCOME COVERED**
**CO1:** *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 4

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**Code of Conduct**
I P. Guru/192521010 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: P.Guru

**Assessment Tasks**
**Concept Mapping Tasks**
Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

**Expected concept nodes:**
- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

**Concept Mapping Rubric**

| Criteria | Weightage | Level 4 - Exemplary | Level 3 - Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Concept Identification | 25% | All major concepts are accurately identified | Most concepts are identified correctly | Limited concepts are identified | Essential concepts are missing |
| Relationship Mapping | 25% | Links between concepts are accurate and meaningful | Most links are correct | Some links are weak or unclear | Relationships are mostly incorrect |
| Hierarchy and Organization | 20% | Excellent structure with clear hierarchy | Mostly organized with minor issues | Basic structure with weak hierarchy | No logical organization |
| Technical Relevance | 20% | Map strongly reflects cloud course content | Mostly aligned to topic | Partially aligned to topic | Weak alignment to topic |
| Visual Clarity | 10% | Neat, readable, and easy to interpret | Generally readable | Somewhat cluttered | Poorly presented |

5. Questions

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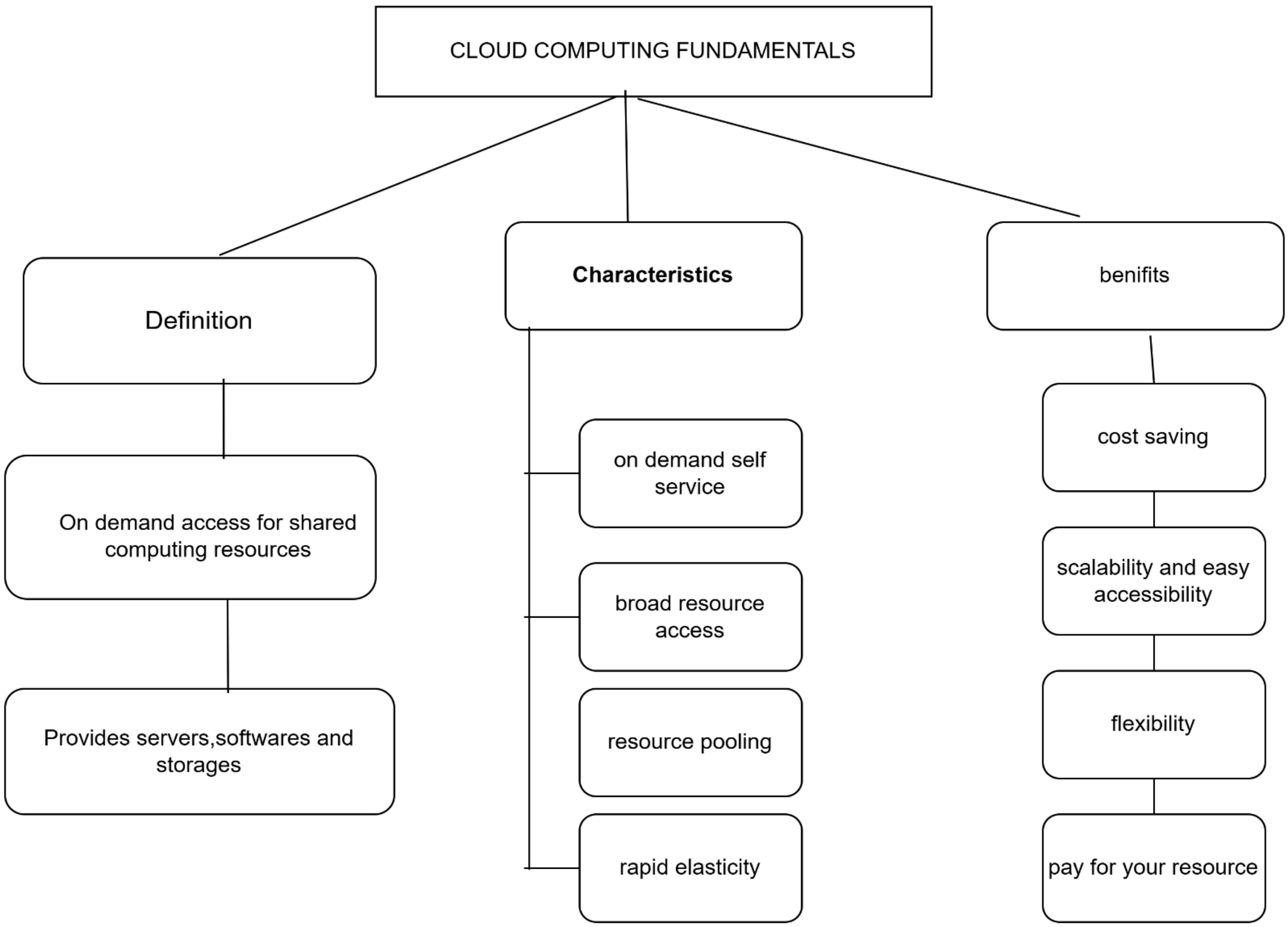
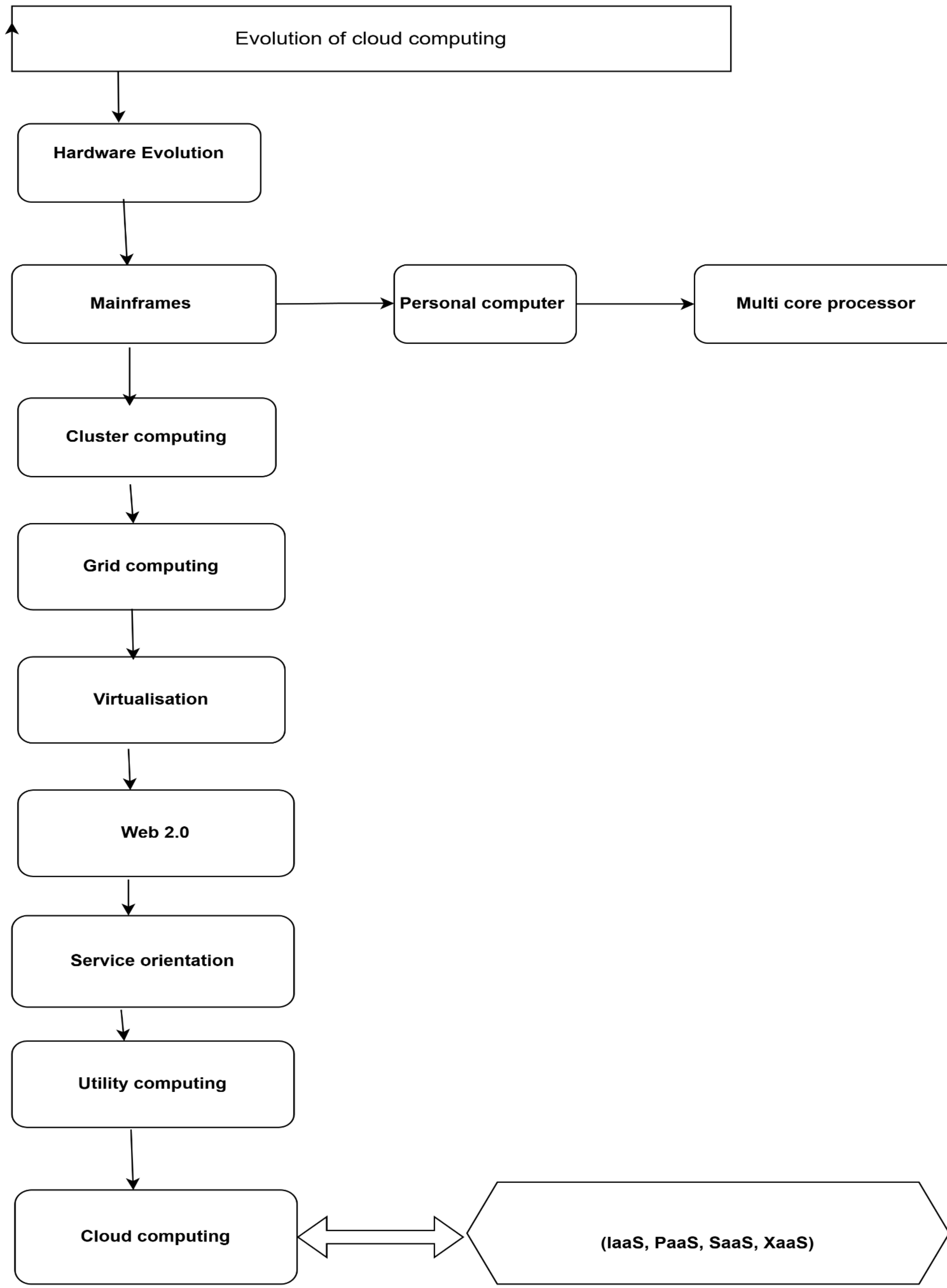
| i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why? |
| ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent? |
| iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system? |
| iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience? |
| v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it? |

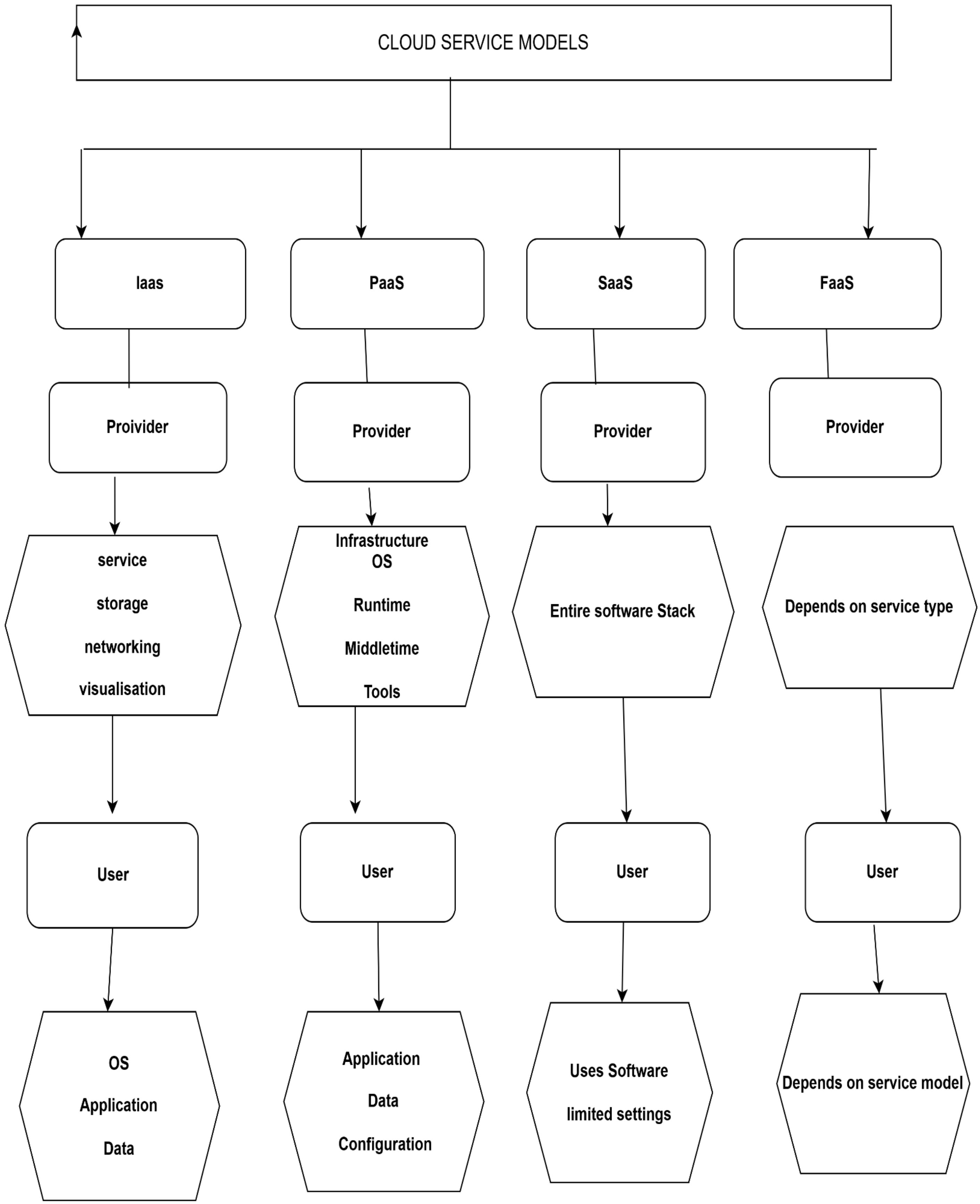
## Concept map 1:



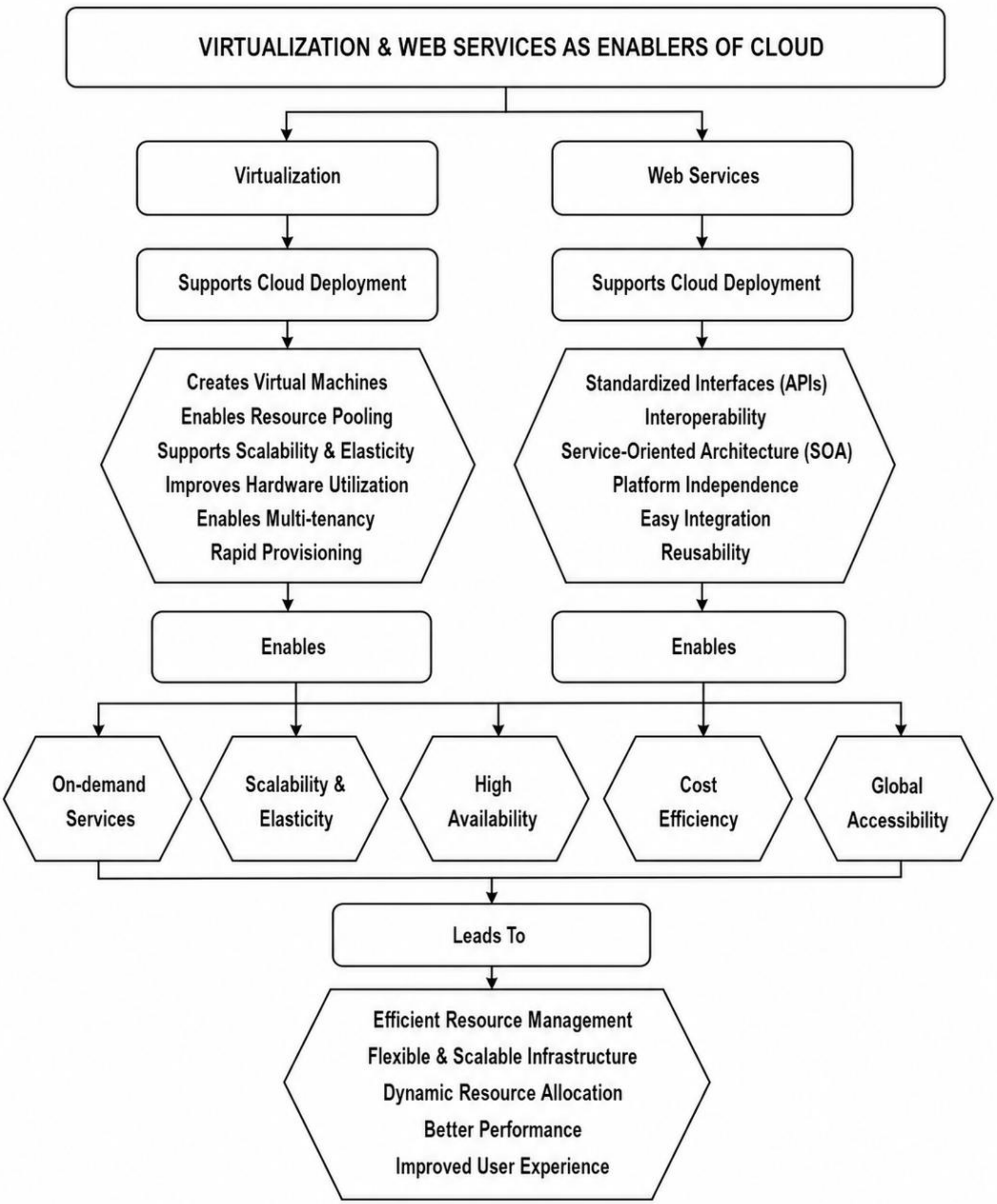
## Concept map 2:



## Concept map 3:



## Concept map 4:



## Answers for Questions:

1) **Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?.**

   The concept map effectively represents cloud computing principles by organizing the concepts into logical sections such as cloud fundamentals, service models, and cloud enablers. The use of hierarchical relationships makes it easier to understand how virtualization and web services support cloud computing. The design decisions that most strongly support the solution are:

   Clear categorization of cloud components.

   Separation of provider and user responsibilities.

   Logical flow showing how cloud technologies enable scalability

   Visual representation of dependencies between concepts.

   These decisions improve readability and help explain complex cloud computing

2) **Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?**

   **Influence of Scalability, Elasticity, and Resource Management**

   Scalability, elasticity, and resource management are essential components of cloud architecture.

   **Scalability** :Allows systems to handle increasing workloads by adding resources when needed.

   **Elasticity** : Enables automatic allocation and deallocation of resources based on demand.

   **Resource management** :Ensures efficient utilization of computing resources such as storage, processing power, and networking.

   If these capabilities were absent:

   - Applications may experience poor performance during high traffic.
   - Resources could remain underutilized or overloaded.
   - Operational costs would increase due to inefficient resource allocation.

- Cloud services would lose their flexibility and reliability.

Thus, these capabilities are fundamental for maintaining efficient and cost-effective **cloud environments.**

3) **Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?**

**Justification of APIs, Web Services, and Cloud Services**

The concept map includes APIs, web services, and cloud services because they are essential for communication and service delivery in cloud computing.

**APIs (Application Programming Interfaces)** enable applications to communicate securely and efficiently.

**Web Services** provide interoperability between different platforms and systems.

**Cloud Services (IaaS, PaaS, SaaS, and XaaS)** offer scalable and flexible solutions for different user requirements.

Their contributions include:

- Improved system integration.
- Faster deployment of applications.
- Reduced infrastructure management costs.
- Better scalability and performance.
- Enhanced accessibility across devices and platforms.

These technologies significantly improve the overall functionality and efficiency of cloud systems.

4) **Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?**

**Relationships Between Components**

The components in the concept map are highly interconnected.

Virtualization supports resource pooling, multi-tenancy, and rapid provisioning.

Web services provide interoperability and seamless integration between applications.

Scalability and elasticity work together to manage workloads efficiently.

Cloud service models distribute responsibilities between providers and users.

These interactions improve:

- **Reliability:** High availability ensures uninterrupted service delivery.
- **Performance:** Dynamic resource allocation optimizes system performance.
- **User Experience:** Users can access services anytime with minimal configuration requirements.
- **Cost Efficiency:** Pay-as-you-go models reduce unnecessary expenditures.

The relationships between these components create a robust cloud ecosystem that is flexible, scalable, and user-friendly.

5) **Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?**

**Reflection on Developing the Concept Map:**

Developing the concept map provided a deeper understanding of how different cloud computing components interact with one another. It highlighted that cloud computing is not merely a collection of services but an integrated architecture involving virtualization, web services, scalability, and resource management.

Through this process, I gained a better understanding of:

- The responsibilities of cloud providers and users.
- The role of virtualization and APIs in enabling cloud services.
- The importance of scalability and elasticity in maintaining performance.